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Python Programming - 2301CS404

Lab - 7

Roll No : 459

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01) WAP to find sum of tuple elements.

Sample Input:

T = (10, 20, 30, 40)

Sample Output:

Sum of tuple elements = 100

```
In [1]: T = (10,20,30,40)
sum = 0
for i in T:
    sum+=i

print("Sum of tuple elements =",sum)
```

Sum of tuple elements = 100

02) WAP to find Maximum and Minimum K elements in a given tuple.

Sample Input:

T = (7, 2, 9, 4, 1, 5)
K = 2

Sample Output:

Minimum 2 elements: (1, 2)
Maximum 2 elements: (9, 7)

```
In [12]: T = (7, 2, 9, 4, 1, 5)
k = 2

T = tuple(sorted(T,reverse=True)) #descending and aa list ma return kare

minTup = (T[-1:-k-1:-1])
maxTup = (T[:k])

print("Minimum 2 elements:",minTup)
print("Maximum 2 elements:",maxTup)
```

Minimum 2 elements: (1, 2)
Maximum 2 elements: (9, 7)

03) WAP to find tuples which have all elements divisible by K from a list of tuples.

Sample Input:

```
tuple_list = [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]
K = 2
```

Sample Output:

Tuples divisible by 2 : [(2, 4, 6), (4, 8, 12)]

```
In [17]: tuple_list = [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]
k = 2

li = []

# for i in range(len(tuple_list)): #we also use without range
#     flag = False
#     for j in tuple_list[i]:
#         if(j % k != 0):
#             flag = True
#             break
#     if(not flag):
#         li.append(tuple_list[i])

# or

for i in tuple_list:

    for j in i:
        if(j % k != 0):
            break
    else:
        li.append(i)

print(li)
```

[(2, 4, 6), (4, 8, 12)]

04) WAP to create a list of tuples from given list having number and its cube in each tuple.

Sample Input:

```
numbers = [1, 2, 3, 4, 5]
```

Sample Output:

List of tuples with number and its cube: [(1, 1), (2, 8), (3, 27), (4, 64), (5, 125)]

```
In [3]: numbers = [1, 2, 3, 4, 5]

t = tuple( (i , i ** 3) for i in numbers )

print(t)
```

((1, 1), (2, 8), (3, 27), (4, 64), (5, 125))

05) WAP to find tuples with all positive elements from the given list of tuples.

Sample Input:

```
tuple_list = [(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3, -4, -5)]
```

Sample Output:

Tuples with all positive elements: [(1, 2, 3), (4, 5, 6)]

```
In [20]: tuple_list = [(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3, -4, -5)]

li = []

for i in tuple_list:

    for j in i:
        if(j <= 0):
            break
    else:
        li.append(i)

print("Tuples with all positive elements:",li)
```

[(1, 2, 3), (4, 5, 6)]

06) WAP to remove tuples of length K.

Sample Input:

```
tuple_list = [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
K = 2
```

Sample Output:

List after removing tuples of length 2 : [(3, 4, 5), (6,), (7, 8, 9)]

```
In [ ]: tuple_list = [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
K = 2

for i in tuple_list:
    if(len(i) == k):
        tuple_list.remove(i)

print("List after removing tuples of length 2 :",tuple_list)
```

List after removing tuples of length 2 : [(3, 4, 5), (6,), (7, 8, 9)]

07) WAP to remove all occurrences of a given element from a tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
x = 3
```

Sample Output:

(5, 7, 9)

```
In [36]: T = (3, 5, 3, 7, 3, 9)
x = 3

# T = list(T)

# T.remove(x) # Aa ekj element remove karse

T = tuple(i for i in T if i != x)

print(T)
```

(5, 7, 9)

08) WAP to remove duplicates from tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
```

Sample Output:

(3, 5, 7, 9)

```
In [25]: T = (3, 5, 3, 7, 3, 9)

T = tuple(set(T))

print(T)
```

(9, 3, 5, 7)

09) WAP to multiply adjacent elements of a tuple and print that resultant tuple.

Sample Input:

t = (2, 3, 4, 5)

Sample Output:

Resultant tuple after multiplying adjacent elements: (6, 12, 20)

```
In [37]: t = (2, 3, 4, 5)
tp = []
for i in range(len(t)-1):
    tp.append(t[i]*t[i+1])

print(tuple(tp))
```

(6, 12, 20)

10) WAP to test if the given tuple is distinct or not.

Sample Input:

tuple1 = (1, 2, 3, 4)
tuple2 = (1, 2, 2, 3)

Sample Output:

(1, 2, 3, 4) is distinct? -> True
(1, 2, 2, 3) is distinct? -> False

```
In [5]: # tuple1 = (1, 2, 3, 4)
tuple1 = (1, 2, 2, 3)

if(len(set(tuple1)) != len(tuple1)):
    print(False)

else:
    print(True)
```

False

11) WAP to rotate the elements of a tuple to the right by k positions.

Sample Input:

```
T = (10, 20, 30, 40, 50)
k = 2
```

Sample Output:

```
(40, 50, 10, 20, 30)
```

```
In [43]: T = (10, 20, 30, 40, 50)
k = 2

tp = []

for i in range(k+1, len(T)):
    tp.append(T[i])

for i in range(k+1):
    tp.append(T[i])

print(tuple(tp))
```

```
(40, 50, 10, 20, 30)
```

12) WAP to merge two tuples of equal length alternately.**Sample Input:**

```
T1 = (1, 3, 5)
T2 = (2, 4, 6)
```

Sample Output:

```
(1, 2, 3, 4, 5, 6)
```

```
In [45]: T1 = (1, 3, 5)
T2 = (2, 4, 6)

tp = []

if(len(T1) == len(T2)):
    for i in range(len(T1)):
        tp.append(T1[i])
        tp.append(T2[i])

print(tuple(tp))
```

```
(1, 2, 3, 4, 5, 6)
```

13) WAP to create a tuple of squares of elements that are even and greater than 10.**Sample Input:**

T = (4, 12, 7, 18, 10)

Sample Output:

(144, 324)

```
In [48]: T = (4, 12, 7, 18, 10)

tp = tuple(i*i for i in T if (i%2==0 and i > 10))

print(tp)

(144, 324)
```

14) WAP to create (index, value) pairs for prime numbers.

Sample Input:

T = (4, 7, 9, 11, 15)

Sample Output:

((1, 7), (3, 11))

```
In [61]: T = (4, 7, 9, 11, 15)

def isPrime(n):
    for i in range(2, int(pow(n, 0.5)+1)):
        if n % i == 0:
            return False
    else:
        return True

tp = []

for i in range(len(T)):
    if(isPrime(T[i])):
        tp.append(tuple([i, T[i]]))

print(tp)
```

[(1, 7), (3, 11)]

15) WAP to split a tuple into even-index and odd-index tuples.

Sample Input:

T = ('a', 'b', 'c', 'd', 'e')

Sample Input:

even = ('a', 'c', 'e')
odd = ('b', 'd')

In [54]: `T = ('a', 'b', 'c', 'd', 'e')`

```
even = []
odd = []

for i in range(len(T)):
    if i % 2 == 0:
        even.append(T[i])
    else:
        odd.append(T[i])
print("even =", even)
print("odd =", odd)
```

even = ['a', 'c', 'e']
odd = ['b', 'd']

16) WAP to compute column-wise sum from a tuple of lists.

Sample Input:

`T = ([1, 2, 3], [4, 5, 6], [7, 8, 9])`

Sample Output:

12 15 18

In [59]: `T = ([1, 2, 3], [4, 5, 6], [7, 8, 9])`
`sm = 0`
`for i in range(len(T)):`
 `for j in range(len(T[0])):`
 `sm += T[j][i]`
`print(sm)`

12
27
45